IS ARCHITECTURAL DIVERSITY NEEDED? SOFT-VOTING ENSEMBLES OF U-NET VARIANTS FOR RETINAL ARTERY–VEIN SEGMENTATION

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ABSTRACT

Separating retinal arteries from veins supports screening for ophthalmic and systemic disease, but single deep networks remain sensitive to the wide range of vessel calibres, low contrast and annotation noise of fundus images. We study whether ensembling helps for this task and, if so, why. On the Fundus-AVSeg dataset, we evaluate U-Net and ResUNet baselines, two extensions (a bi-directional feature pyramid neck and a morphological top-hat input block) and their transfer to the ResUNet backbone. Averaging the softmax outputs of three models significantly outperforms every single model over four training seeds (mean artery/vein Dice 0.824 ± 0.006 vs. at most 0.813) and reaches 0.838 in our best single-seed configuration. An ensemble of three identical, independently seeded networks recovers about 80% of this gain, indicating that it stems mainly from averaging out training variability rather than from architectural diversity. Soft voting thus offers a simple, parameter-free improvement that does not require designing complementary architectures.

***Index Terms*—** Retinal imaging, artery–vein segmentation, U-Net, ensemble learning, deep ensembles

1. INTRODUCTION

The retinal vasculature is the only part of the circulatory system observable non-invasively, and arteriolar narrowing and venular widening, summarised for instance by the arteriovenous ratio, are established biomarkers of ophthalmic and systemic disease [1, 2]. Separating arteries from veins is therefore clinically important, but manual A/V annotation is slow and subject to observer variability. Automated methods range from fully convolutional A/V networks [3] to uncertainty-aware classification [4], and are dominated by the U-Net [5] and its descendants [6, 7, 8], which still struggle with the wide scale range of fundus vessels, low contrast that fragments thin vessels, and annotation noise [2, 9].

Two directions address these issues within a single network: feature-pyramid necks such as BiFPN [10], adapted to fundus vessels in [2], which fuse multi-scale features through learnable weighted connections; and morphological filtering, which injects classical priors for thin tubular structures [9]. A third, architecture-agnostic option is ensembling: averaging class posteriors (the *sum rule*) is among the most robust combination schemes [11, 12], and ensembles of independently trained deep networks are known to improve accuracy even when all members share one architecture [13, 14].

For A/V segmentation, it remains unclear whether combining architecturally different models helps, and whether any gain comes from their complementarity or simply from averaging independently trained networks. If diversity is not needed, design effort is better spent on a single strong model.

Contributions. (1) On Fundus-AVSeg [1], we evaluate U-Net, ResUNet, BiFPN-UNet and Morpho-U-Net, and transfer both extensions to the ResUNet backbone. (2) We show over four seeds that a parameter-free soft-voting ensemble significantly outperforms every single model. (3) With a same-architecture control, we show that most of the gain comes from averaging independently trained networks, confirming in the A/V setting what has been reported for deep ensembles [13, 14].

2. METHODS

Dataset. Fundus-AVSeg [1] contains 100 colour fundus images (1280^2 or 2656×1992 px) with pixel-wise A/V annotation by ophthalmologists, split into 70/10/20 training/validation/test images. Crossing and uncertain pixels (together $<3\%$ of vessel pixels) are too scarce to be learned as separate classes and are mapped to background, giving three classes. All images are resized to 592×592 without padding or cropping (bilinear interpolation for images, nearest-neighbour for masks).

Baselines. A standard U-Net [5] with six resolution levels and a ResUNet [8] in which every block is a residual block (conv–BN–ReLU–conv–BN with a 1×1 projection shortcut; channels $64 \rightarrow 1024$, 32.4M parameters). ResUNet is the strongest baseline reported by the dataset authors, so we reproduce it in our setting.

BiFPN-UNet. A weighted bi-directional feature-pyramid neck is inserted between the U-Net encoder and decoder. The encoder features $P_1^{\text{in}}, \ldots, P_6^{\text{in}}$ are fused along a top-down and a bottom-up path; each node combines its inputs I_i by fast

normalised fusion [10]

$$O = \sum_i \frac{w_i}{\epsilon + \sum_j w_j} I_i, \qquad w_i \geq 0,\ \epsilon = 10^{-3}, \quad (1)$$

followed by a convolution, so that each scale learns how to trade off high-resolution (localisation) and low-resolution (semantic) inputs. We evaluate a 64–1024 channel and a lean 32–256 channel variant. **Morpho-U-Net.** A parameter-free preprocessing block adds classical morphological descriptors of small-scale structure to the input. After Gaussian smoothing (5×5, $\sigma = 1$), we compute per RGB channel a white top-hat (image minus its 3×3 opening), which responds to fine structures brighter than their immediate surroundings, and an erosion-depth map (sum of 10 successive 3×3 grey-level erosions), a grey-level analogue of the distance transform. Both are concatenated with the input, giving a 9-channel input to the U-Net encoder.

ResUNet-backbone variants. To test whether both extensions transfer to the stronger backbone, *BiFPN-ResUNet* inserts the BiFPN neck between the ResUNet encoder and decoder, and *Morpho-ResUNet* feeds the morphological input to the ResUNet encoder; both reuse the ResUNet training recipe.

Soft-voting ensemble. Given K models producing per-pixel softmax vectors $\mathbf{p}^{(k)}(x)$ over {bg, artery, vein}, the ensemble predicts

$$\bar{\mathbf{p}}(x) = \frac{1}{K} \sum_{k=1}^{K} \mathbf{p}^{(k)}(x), \qquad \hat{y}(x) = \arg\max_c \bar{p}_c(x). \quad (2)$$

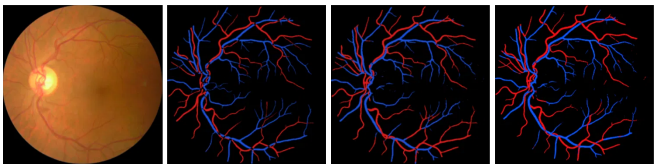
Averaging distributions rather than hard masks preserves each model's confidence, adds no parameters and needs no threshold. We use $K = 3$ with uniform weights over three members: **A** BiFPN-UNet (64–1024ch), **B** U-Net and **C** Morpho-U-Net. We also replace C by ResUNet-backbone models, and build a *same-architecture control* of three independently seeded BiFPN-UNets to isolate averaging from architectural diversity.

3. EXPERIMENTS

Implementation. All models are trained in PyTorch on an NVIDIA Quadro RTX 8000. BiFPN, Morpho and ResUNet variants use AdamW (lr 5×10^{-4}, weight decay 2×10^{-4}, cosine annealing), batch size 4, up to 150 epochs (300 for BiFPN-UNet 64–1024ch and ResUNet), the loss $\mathcal{L} = 0.7\,\mathcal{L}_{\text{CE}} + 0.3\,\mathcal{L}_{\text{Dice}}$ with inverse-frequency class weights, and flip/$\pm 10°$ rotation augmentation. The U-Net baseline follows a simpler recipe (Adam, lr 10^{-3}, batch size 2, 150 epochs, weighted cross-entropy, no augmentation); as it reproduces the published U-Net score of [1] (Table 1), we keep it as a reference, but differences between it and the other models reflect both architecture and training recipe. For every model, the checkpoint with the best test Dice was retained; this rule, applied to all single models and ensemble members alike, does not favour the ensemble but makes absolute scores optimistic, so we base our conclusions on paired, multi-seed comparisons.

Table 1. A/V Dice/IoU on Fundus-AVSeg. Top: ours (fixed 592^2 resolution, single split, seed 42). Bottom: results reported by [1] (native-resolution patches, 5-fold CV; mean of their artery and vein columns), shown for reference only, as the protocols differ. A: BiFPN-UNet (64–1024ch), B: U-Net, C: Morpho-U-Net.

Model	Dice	IoU
This work: single models		
BiFPN-UNet (32–256ch)	0.7906	0.6557
Morpho-U-Net (top-hat + erosion)	0.7917	0.6572
U-Net (baseline)	0.8023	0.6719
Morpho-ResUNet	0.8070	0.6781
BiFPN-UNet (64–1024ch)	0.8092	0.6812
ResUNet	0.8103	0.6828
BiFPN-ResUNet	**0.8147**	**0.6889**
This work: soft-voting ensembles		
A+B (two members)	0.8296	0.7106
A+B+C (Morpho-U-Net)	0.8320	0.7139
A+B+ResUNet	0.8350	0.7184
A+B+Morpho-ResUNet	0.8364	0.7203
A+B+BiFPN-ResUNet	**0.8380**	**0.7228**
Reported by Deng et al. [1] (different protocol)		
U-Net	0.8027	0.6707
ResUNet (best reported)	0.8319	0.7126



(a) Input (b) Ground truth (c) ResUNet (d) Ensemble

Fig. 1. Qualitative A/V segmentation on test image 092_N (arteries in red, veins in blue). (d): A+B+BiFPN-ResUNet ensemble. Test-set mean A/V Dice: ResUNet 0.8103, ensemble 0.8380.

Metrics and statistics. We report *A/V Dice* and *A/V IoU*, i.e. the mean of the artery and vein scores (background excluded), averaged over the 20 test images, matching the per-class protocol of [1]. Ensemble–model comparisons use one-tailed Wilcoxon signed-rank tests paired over test images, with Holm correction within each table; 95% CIs are obtained by bootstrap resampling of test images.

3.1. Main results

Table 1 summarises all single models and ensembles at the primary seed, with the results reported by the dataset authors

Table 2. A/V Dice of the four original architectures and the A+B+C ensemble over four seeds (42, 1, 7, 13): mean, standard deviation, bootstrap 95% CI, and one-tailed Wilcoxon p (ensemble $>$ model, before Holm correction).

Model	Mean	Std	95% CI	p
Morpho-U-Net	0.8017	0.0070	[0.783, 0.817]	<1e-4
U-Net	0.8082	0.0045	[0.791, 0.823]	<1e-4
BiFPN-UNet	0.8077	0.0037	[0.792, 0.823]	<1e-4
ResUNet	0.8129	0.0022	[0.796, 0.827]	<1e-4
Ensemble (A+B+C)	**0.8242**	**0.0056**	**[0.809, 0.839]**	–

for reference.

Single models. The ResUNet backbone is the strongest, and both extensions can be transferred to it. BiFPN-ResUNet is the best single model (Dice 0.8147, IoU 0.6889), $+0.44$ points above plain ResUNet (0.8103). Morpho-ResUNet (0.8070) improves on Morpho-U-Net by 1.5 points, though it trails plain ResUNet. The wide BiFPN-UNet outperforms the lean variant (0.8092 vs. 0.7906); note that the wide variant was also trained for longer.

Ensembles. At this seed, every ensemble beats every single model. The A+B+C ensemble reaches 0.8320 Dice, and using a ResUNet-backbone model as the third member raises this further, up to 0.8380 Dice / 0.7228 IoU for A+B+BiFPN-ResUNet (Sec. 3.4).

Relation to [1]. Our U-Net reproduces the published figure (0.8023 vs. 0.8027), whereas our ResUNet is 2.2 points below its patch-based, cross-validated counterpart, which residual networks appear to benefit from. The figures are thus not directly comparable; our ensembles reach the level of the best published single model (0.8319).

Qualitative results. In Fig. 1, ResUNet and the best ensemble both recover the main arcades and label the large vessels correctly; they differ mainly in thin, low-contrast peripheral branches, where the errors of all models concentrate (Sec. 3.5).

3.2. Statistical significance

As the test set has only 20 images, we retrained the four original architectures with three extra seeds (1, 7, 13) and rebuilt the A+B+C ensemble at each seed (Table 2). Over seeds, the ensemble reaches 0.8242 ± 0.0056 Dice, a more conservative estimate than the primary run (0.8320) but still 1.1 and 1.6 points above the mean ResUNet and U-Net, and it significantly outperforms every single model (all Holm-adjusted $p < 10^{-3}$). Its seed-to-seed spread is larger than that of most single models; with four seeds such estimates are themselves uncertain, and the spread is driven largely by the high primary-run value.

3.3. Ablations

Ensemble composition. The two-member ensemble of BiFPN-UNet and U-Net already reaches 0.8296 Dice; adding Morpho-U-Net brings a further $+0.0024$, smaller than the ensemble's seed-to-seed spread, so the two-member ensemble is a cheaper alternative.

Morphological features. On the U-Net backbone (seed 42), using only the white top-hat (6 channels) gives the best morphological variant (Dice 0.8111), above the full 9-channel input (0.7917) and the only one above the U-Net baseline (0.8023). The erosion-depth map brings no gain alone (0.8017) and degrades results when combined; replacing the white by the black top-hat also lowers Dice (0.7996). As vessels are darker than the background, the advantage of the white over the black top-hat suggests that, at this 3×3 scale, the filter mainly captures fine bright detail such as the central light reflex, more pronounced in arteries, i.e. an artery/vein cue rather than a detection cue.

3.4. Ensembling ResUNet-backbone models

We replace the weakest member of the A+B+C ensemble (C, Morpho-U-Net) with each ResUNet-backbone model, keeping A (BiFPN-UNet) and B (U-Net) (Table 1). All three substitutions improve significantly on A+B+C (one-tailed Wilcoxon, $n = 20$: $p < 10^{-4}$, 0.0004 and 0.0164 for BiFPN-ResUNet, Morpho-ResUNet and ResUNet), also after Holm correction. A+B+BiFPN-ResUNet reaches 0.8380 Dice / 0.7228 IoU, $+2.3$ points above its best single member. The gain tracks member strength only loosely: Morpho-ResUNet is weaker alone than plain ResUNet (0.8070 vs. 0.8103) yet yields a slightly better ensemble. As these results are single-seed, they should be read as upper estimates (cf. Sec. 3.2).

3.5. Why does the ensemble help?

The models largely agree: they disagree on 1.9–3.1% of pixels, and Yule's Q [15] between their errors lies between 0.997 and 0.999 for arteries and veins. These all-pixel figures are dominated by easy background; the remaining errors concentrate on thin, low-contrast and crossing vessels, where members are individually uncertain and probability averaging can correct them. The decisive test is the same-architecture control: three independently seeded BiFPN-UNets reach 0.8203 mean Dice (95% CI $[0.804, 0.835]$), i.e. $+1.3$ points over the mean single BiFPN-UNet, compared with $+1.6$ points for the heterogeneous A+B+C ensemble over its best member. Averaging identical architectures thus recovers about 80% of the gain, and the remaining 0.0039 difference is smaller than the seed-to-seed spread. We therefore attribute most of the gain to averaging out the variability of independently trained networks; architectural diversity adds at most a small increment, and architecture matters chiefly through the strength of the members.

4. CONCLUSION

On Fundus-AVSeg, a parameter-free soft-voting ensemble significantly outperforms every single model over four seeds (0.8242 ± 0.0056 A/V Dice), and stronger ResUNet-backbone members raise it to 0.838 at the primary seed. A same-architecture control recovers most of this gain, which therefore comes mainly from averaging independently trained networks: averaging a few strong models does not require designing complementary architectures. Among single models, BiFPN-ResUNet performs best, and morphological priors help most in a white-top-hat-only form. Limitations include a single-site, 100-image dataset; checkpoint selection on the test set, applied uniformly to all models; a simpler training recipe for the U-Net baseline; single-seed results for the ResUNet-backbone variants; and the threefold inference cost of the ensemble. Future work will validate the ResUNet-backbone ensembles across seeds and on the GAVE2 challenge benchmark, and distill the ensemble into a single model.

5. COMPLIANCE WITH ETHICAL STANDARDS

This research study was conducted retrospectively using human subject data made available in open access by Deng et al. [1]. Ethical approval was not required as confirmed by the license attached with the open access data.

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